SIMATS SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

Engineer to Excel

PLACEMENT & TRAINING CELL

Problem Statements-SET 1

1. Write a program to draw Archimedean spiral.(TUTRLE)



2. Write a program that takes four parameters a, b, c and n—and checks to see if Fermat's theorem holds. If n is greater than 2 and $a^n + b^n = c^n$ the program should print, "Holy smokes, Fermat was wrong!" Otherwise the program should print, "No, that doesn't work."

   Enter A:0
   Enter B:1
   Enter n:3

   Sample Output:
   Holy Smokes, Fermat was wrong.

   Enter A:1
   Enter B:1
   Enter n:3

   Sample Output:
   No that doesn't work.

3. Given an infinite supply of each denomination of Indian currency [1, 2, 5, 10, 20, 50, 100, 200, 500, 2000] and a target value N.
   Find the minimum number of coins and/or notes needed to make the change for Rs N.
   You must return the list containing the value of coins required

   Input 44

   Output: 20 20 2 2

   Test Cases:

Problem Statements-SET 1

Input 221.50, 10.10, 1,00,000 -560.00

4. Given an integer N denoting the Length of a line segment. You need to cut the line segment in such a way that the cut length of a line segment each time is either x , y or z. Here x, y, and z are integers.

   After performing all the cut operations, your total number of cut segments must be maximum.

   Sample Input
   N=5
   X=5
   Y=3
   Z=2

   Sample Output:2

   Test cases:
   N=6.0, N=5.1, N=5.001, N=010110

5. Given an array arr of size N containing non-negative integers, the task is to divide it into two sets S1 and S2 such that the absolute difference between their sums is minimum and find the minimum difference

   Input: N=4

   arr[]={1,6,11,5}

   Exp: Subset1={1,5,6} Sum of Subset 1=12

   Subset2={11} Sum of Subset2=11

   Difference is 1.

6. Given a sorted array arr[] of size N. Find the element that appears only once in the array. All other elements appear exactly twice.

   Sample Input and Output

N=11

arr=[1,1,2,2,3,3,4,50,50,65,65]

Output: 4

Test Cases:
N=1, arr=[0]
N=1.0 arr=[0.0]
N=0
N=1% arr=['e']

7. Given a non-negative integer N. The task is to check if N is a power of 2. More formally, check if N can be expressed as 2^x for some x.

   (Hint: Your task is to write a function which takes N as a parameter and returns true or false by checking if the given number can be represented as a power of two or not.)

   Input N=1

   Output: Yes

   Test Cases N=1.1, 1.0000, 1,616, 1$

8. Given non-zero two integers N and M. The problem is to find the number closest to N and divisible by M. If there are more than one such number, then output the one having maximum absolute value

   Input

   N=13

   M=4

   Output

   The closest divisible integer is 12.0

   Test Cases:

   N=13.0, M=4.1

   N=1,00,000 M=1,000

Problem Statements-SET 1

N=500%, M=5%

N=-100, M=-12

9.Design a system that takes big URLs like "http://www.amazon.com/count-sum-of-digits-in-numbers-from-1-to-n/" and converts them into a short URL. It is given that URLs are stored in database and every URL has an associated integer id. So your program should take an integer id and generate a URL.

Hint:

A URL character can be one of the following

- A lower case alphabet ['a' to 'z'], total 26 characters
- An upper case alphabet ['A' to 'Z'], total 26 characters
- A digit ['0′ to '9'], total 10 characters
- There are total 26 + 26 + 10 = 62 possible characters.

Input

Enter your Id: 12345

Output: Short URL is tgaht

Test Cases:

Enter your id: 1.12345, 0.12333, 000001, 1,00,000

10. You have given a number N. You need to print the pattern for the given value of N.

Hint:

For N = 2 the pattern will be

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Problem Statements-SET 1

2 2 1 1

2 1

Sample Input

N = 3

Output:

3 3 3 2 2 2 1 1 1

3 3 2 2 1 1

3 2 1